Bahir Dar University — Bahir Dar Institute of Technology
Faculty of Computing — MSc in Information Technology

Annotation Guideline

Multi-Task Labeling for Sentiment, Emotion, and Fine-Grained Hate Speech Detection in Amharic-English Code-Switched Social Media Text

1. Purpose and Scope

This guideline instructs annotators on how to assign **Sentiment**, **Emotion**, and **Fine-Grained Hate Speech** labels to social media comments collected from Facebook, Telegram, X, and YouTube for the thesis "**An Explainable Multi-Task Transformer Framework with Parameter-Efficient Fine-Tuning for Sentiment, Emotion, and Fine-Grained Hate Speech Detection in Amharic-English Code-Switched Social Media Text".**

Each comment receives three independent label sets and must be evaluated as a whole — meaning, tone, and intent take priority over any single word or phrase.

1.1 **Annotation Objective:** Annotators will encounter four text varieties, all of which follow the same labeling rules and **The goal is to assign labels to social media comments written in:**

Variety	Example
Pure Amharic (Ge'ez script)	**ዋው በጣም ደስ ያለ ፊልም ነበር።**
Pure English	Wow, this movie was really enjoyable.
Code-mixed (Ge'ez + English)	**ዋው በጣም ደስ ያለ** film **ነበር።**
Romanized Amharic (Latin script)	Waw betam des yale film neber.

2. General Annotation Principle

- Read the entire comment before labeling — never decide from a single word.
- Combine the meaning of Amharic and English when the text is mixed; label the overall message, not each language separately.
- Judge intended meaning, not literal wording — this matters most for sarcasm and idioms.
- Do not annotate emoji-only comments, repeated symbols, bare links, usernames, or numbers without semantic content — mark these "Not Annotatable" and skip.
- If the meaning is genuinely ambiguous after careful reading, flag the comment for adjudication rather than guessing.
- When multiple meanings exist, select the label that best reflects the overall message.

"good actor but the movie is boring" → **Sentiment: Negative (overall opinion dominates the isolated positive word)**

3. Task 1 Sentiment Annotation

Assign exactly ONE label per comment.

Label	Assign when the comment expresses…	Example → Label
Positive	Happiness, appreciation, love, satisfaction, support, praise	"betam konjo music new" → Positive
Negative	Dislike, complaint, anger, disappointment, criticism	"**መጥፎ ስራ ነው**" → Negative
Neutral	Pure information; no clear positive or negative stance	"The movie was released in 2025" → Neutral

4. Task 2 — Emotion Annotation

Select ONE OR MORE labels per comment (multi-label). Choose Neutral only when no other emotion applies.

Label	Definition	Example → Label
Joy	Happiness, excitement, pleasure	"**በጣም ደስ ብሎኛል**" → Joy

Love	Affection, admiration, attachment	"I love you teddy" → Love
Anger	Frustration, hatred, irritation	"I hate this situation" → Anger
Sadness	Sorrow, loss, disappointment	"I miss my father" → Sadness
Fear	Worry, anxiety, sense of danger	"I am afraid of this" → Fear
Disgust	Strong dislike or rejection	"This is disgusting" → Disgust
Surprise	Shock or unexpected reaction	"Wow, I cannot believe this" → Surprise
Neutral	No identifiable emotional content	"The video duration is 5 minutes" → Neutral

"I love this song but I miss my father" → **Emotion: Love, Sadness (both co-occur; label all that apply)**

5. Task 3 — Fine-Grained Hate Speech Annotation

Assign exactly ONE label per comment. This label set matches the four hate-speech classes in the thesis concept note.

Label	Definition	Example → Label
Normal	Ordinary conversation, opinion, or discussion; no attack on any person or group	"I love Ethiopian music" → Normal
Offensive	Insult or rude expression directed at a specific individual (not a group)	"You are stupid" → Offensive
Hate	Attack or discrimination against a group based on ethnicity, religion, nationality, gender, or another protected identity	"All people from this group are bad" → Hate
Threat	Explicit statement of intent to cause harm	"I will kill you" → Threat

5.1 Target-Group Identity Categories (for the Hate label)

A comment qualifies as Hate only if the attack targets a group identity, including but not limited to: ethnicity, religion, nationality, gender, political affiliation as a group identity, disability, or other protected characteristics. An attack on one named individual, regardless of severity, is Offensive or Threat — not Hate.

5.2 Religious and Political Discussion

Expressing or defending a belief is Normal. Labeling as Hate requires an explicit attack on a group.

"I support my religion" → **Hate: Normal**

"Followers of this religion are evil" → **Hate: Hate**

6. Special Handling Rules

6.1 Mixed Sentiment

When a comment contains both positive and negative content, label the dominant or concluding opinion.

"The actor is good but the story is boring" → **Sentiment: Negative**

6.2 Sarcasm and Irony

Label the intended meaning, not the literal words.

> *"Amazing job destroying everything"* → **Sentiment: Negative**

6.3 Romanized Amharic and Code-Switching

Read romanized Amharic phonetically and interpret it exactly as its Ge'ez-script equivalent would be interpreted. In code-mixed text, base the label on the combined meaning of both languages rather than defaulting to whichever language appears first or more frequently.

6.4 Non-Annotatable Content

Skip and tag as "Not Annotatable": emoji-only comments, repeated punctuation or symbols, bare URLs, @usernames with no accompanying text, and numeric strings with no semantic content.

6.5 Unclear or Borderline Cases

Do not guess. Flag the comment with a short note explaining the ambiguity and route it to the adjudication step described in Section 8.

7. Annotation Decision Order

Follow this sequence for every comment:

- Step 1 — Read and understand the complete text (all languages present).
- Step 2 — Assign Sentiment: Positive / Negative / Neutral (choose one).
- Step 3 — Assign Emotion: one or more of Joy, Love, Anger, Sadness, Fear, Disgust, Surprise, Neutral.
- Step 4 — Assign Hate Speech category: Normal / Offensive / Hate / Threat (choose one).
- Step 5 — If any label was uncertain, add a one-line justification note in the remarks column.

7.1 Final Annotation Table Format

Text	Sentiment	Emotion	Hate
betam konjo music new	Positive	Joy	Normal
I miss my father	Negative	Sadness	Normal
You are stupid	Negative	Anger	Offensive
I will kill you	Negative	Anger	Threat

8. Quality Control

- Every comment is labeled independently by at least two annotators before any comparison is made.
- A calibration round of 50 comments is annotated by all annotators before full-scale annotation begins, followed by a joint discussion of disagreements.
- Disagreements are resolved by a third senior annotator (adjudicator) whose decision is final and is recorded with a short rationale.
- Weekly spot-checks on 5% of completed batches are conducted to catch labeling drift.

9. Annotation Tool

- All annotation is carried out in the project's configured Label Studio instance, using the labeling interface set up for the three tasks (sentiment, emotion, hate speech).

10. Annotator Quality Checklist

Confirm all items before submitting a batch:

- I read the whole comment before assigning any label.
- I considered Amharic and English content together, including romanized Amharic.
- Sentiment has exactly one label.
- Emotion has one or more labels, with Neutral used only when no other emotion applies.
- Hate Speech has exactly one label, and Hate is reserved for group-directed attacks.
- I labeled overall meaning, not a single word or phrase in isolation.
- I interpreted sarcasm by its intended meaning.
- I flagged unclear or borderline comments for adjudication instead of guessing.
- I skipped non-meaningful content (emoji-only, links, usernames, symbols).